



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai 2026



ASSESSMENT TOOL -1

C02_AT1

Name : P Giriprasad Reddy

Register No : 192525172

Course code : CSA1613

Course name: Data Warehousing and Data Mining

Faculty name: Dr Kanimozhi K.V

SIMATS ENGINEERING ASSESSMENT TOOL 1

CO2: Data Preprocessing

Course Code: CSA1613

Course Name: Data Warehousing and Data Mining

Name: P Giriprasad Reddy

Reg No:192525172

Q1. Covariance Analysis in Health Insurance

Data:

x_i (Premium Increase %)	y_i (Claims per 100)
3.2	15
4.0	18
5.5	25
4.8	20

Step 1: Means

$$\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 17.5 / 4 = 4.375$$

$$\bar{y} = (15 + 18 + 25 + 20) / 4 = 78 / 4 = 19.5$$

Step 2: Deviations and Products

$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
-1.175	-4.5	5.2875
-0.375	-1.5	0.5625
1.125	5.5	6.1875
0.425	0.5	0.2125

Sum of products = 12.25

Step 3: Covariance

$$\text{Cov}(x,y) = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / n = 12.25 / 4 = 3.0625$$

Interpretation: Since $\text{Cov}(x,y) > 0$, premium increase rate and claim frequency have a POSITIVE relationship — as the premium increase rate rises, the number of claims per 100 policyholders also tends to rise.

Q2. Data Smoothing using Equal-Frequency Binning

Sorted Data (27 values): 110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Using bin depth = 3 (9 equal-frequency bins):

Bin	Values	Bin Mean (Smoothed)
B1	110, 112, 115	112.33
B2	115, 118, 120	117.67
B3	120, 122, 124	122.00
B4	124, 126, 126	125.33
B5	126, 126, 130	127.33
B6	135, 135, 138	136.00
B7	138, 138, 138	138.00
B8	140, 145, 150	145.00
B9	152, 160, 180	164.00

Q3. Equal-Width Binning for Risk Categorization

Sorted BMI Data (18 values): 18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Range = $39.5 - 18.5 = 21$. Using 3 equal-width bins $\rightarrow$ Width = $21 / 3 = 7$

Bin	Boundaries	Values
B1	18.5 – 25.5	18.5, 19.2, 21.8, 24.9
B2	25.5 – 32.5	25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9
B3	32.5 – 39.5	32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Smoothing by nearest bin boundary:

Original Value	Nearest Boundary
----------------	------------------

18.5	18.5
19.2	18.5
21.8	18.5
24.9	25.5
25.5	25.5
26.2	25.5
26.4	25.5
27.8	25.5
29.2	32.5
30.2	32.5
30.4	32.5
31.9	32.5
32.4	32.5
33.1	32.5
33.6	32.5
34.7	32.5
38.2	39.5
39.5	39.5

Each BMI value is replaced with whichever of its bin's two boundaries is closer.

Q4. Data Reduction using Bin Medians

Sorted Cost Data in ₹ (27 values): 120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Using bin depth = 3 (9 bins), each value is replaced by the bin median:

Bin	Values	Bin Median
B1	120, 140, 145	140
B2	145, 148, 150	148
B3	150, 150, 155	150
B4	155, 155, 155	155
B5	160, 160, 165	160
B6	170, 170, 175	170

B7	175, 175, 175	175
B8	180, 190, 200	190
B9	210, 230, 300	230

This reduces 27 raw cost values to 9 representative bin-median values, cutting data volume while preserving the price distribution's shape.

Q5. Standardization of Patient Age Data

Data: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28 (n = 10)

Step 1: Mean

Sum = $18+22+25+42+28+43+33+35+56+28 = 330$

Mean ($\bar{x}$) = $330 / 10 = 33$

(a) Mean Absolute Deviation (MAD)

x_i	$ x_i - 33 $
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5

Sum of absolute deviations = 88

MAD = $88 / 10 = 8.8$

(b) Z-scores for the First Four Observations

Formula: $z = (x_i - \bar{x}) / \text{MAD}$

x_i	Calculation	z-score
18	$(18 - 33) / 8.8$	-1.7045
22	$(22 - 33) / 8.8$	-1.2500

25	$(25 - 33) / 8.8$	-0.9091
42	$(42 - 33) / 8.8$	+1.0227

Interpretation: 18 lies about 1.70 MAD units below the mean age, while 42 lies about 1.02 MAD units above it — both flag moderate deviations from the group's central age.